

Darsh Soni

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EDUCATION

• Vellore Institute of Technology, Chennai

B.Tech Computer Science Engineering (AI & Robotics)

Aug 2024 - June 2028

CGPA : 8.38/10

WORK EXPERIENCE

• Centre of Cyber Security and Physical Systems, VIT Chennai | AI Research Intern

May 2026 - Aug 2026

- **Glacier Displacement Pipeline** : Developing a multi-notebook glacier displacement estimation, velocity mapping, and prediction pipeline for the Daugaard-Jensen Glacier, East Greenland, using Sentinel-1 SAR imagery, ESA SNAP, DInSAR processing, Google Earth Engine, Python, and ML-based forecasting models.
- Building preprocessing, velocity extraction, and temporal analysis workflows for horizontal and vertical glacier displacement visualization using Google Earth Engine and Python, enabling systematic spatio-temporal forecasting.

• BSERC, IIT Madras | Space Tech Intern

Jan 2026 - Mar 2026

- **Drone & GIS Systems** : Worked on drone systems, PCB prototyping, GIS workflows, and remote sensing applications for advanced aerospace research.
- Studied CanSat/CubeSat systems, telemetry, and rocketry technologies, applying the learnings to OpenVSP-based aircraft modeling documentation.

• Google Student Ambassador

Aug 2025 - Dec 2025

- Served as Google's sole student liaison for 34,000+ students at VIT Chennai, conducting GenAI workshops to drive AI literacy and adoption across campus.

• GirlScript Summer of Code (GSSoC) | Open Source Contributor

Jul 2025 - Oct 2025

- Contributed 24+ merged pull requests across open-source web and software projects, placing in the top 1% of contributors.

• French Club, VIT Chennai | Team Lead

Jul 2025 - Oct 2025

- Led planning and coordination of club events, competitions, and international workshops, strengthening cross-cultural engagement on campus.

PROJECTS

• Glacier Velocity Estimation and Spatio-Temporal Forecasting using Sentinel-1 SAR

Research (Ongoing)

- Developed a Sentinel-1 SAR and ESA SNAP based glacier monitoring pipeline to generate a multi-year velocity dataset comprising 17 Sentinel-1 derived velocity maps (2017–2024), 16 temporal forecasting pairs, and 480 glacier velocity patches for spatio-temporal glacier velocity prediction.
- Designed and evaluated CNN-based glacier velocity forecasting models with five-fold cross-validation, and currently developing ConvLSTM and Swin Transformer architectures for long-term spatio-temporal glacier velocity prediction.

• Multi-Stage NL to ER Diagram & SQL Generation System

Research (Submitted Under IEEE Review)

- Developed an agentic AI framework using Google ADK to convert natural language into ER diagrams and multi-dialect SQL schemas, including Oracle SQL, MySQL, SQLite, and PostgreSQL.
- Implemented multi-stage, multi-agent workflows involving ambiguity detection, schema refinement, ER generation, and SQL generation, deployed as a live web-based tool.

• AI Based Railway Crack & Obstacle Detection System

Patent Application Filed (Under Examination)

- Designed and developed a railway safety prototype using MPU6050, IR, and ultrasonic sensors for real-time railway crack, obstacle, and vibration detection.
- Developed computer vision models for railway crack detection and obstacle-type classification using OpenCV and YOLO, contributing to the patent application and intelligent railway monitoring system design.

• Darsh Archive: Personal Portfolio

SvelteKit, MERN, UI/UX Design

- Designed and developed a fully responsive, retro-inspired developer portfolio using the SvelteKit framework, deployed live on GitHub Pages.
- Implemented dynamic routing, multilingual support, custom UI components, and interactive UI elements.

SKILLS

• **Programming Languages** : Python | C++ | C | Java | JavaScript | SQL

• **Libraries & Frameworks** : NumPy | Pandas | OpenCV | PyTorch | CNN | ConvLSTM | Swin Transformer | React | Node.js | Express.js | SvelteKit | REST APIs | Firebase

• **Tools/Utilities** : Git | Docker | MATLAB | Oracle SQL | MySQL | MongoDB | Google Earth Engine | ESA SNAP | DInSAR | Embedded Systems

ACHIEVEMENTS

• Secured 3rd place among 180+ teams in the TechnoVIT CloudOps Hackathon.

• Secured 2nd place among 50+ teams in the MediCode Hackathon.

• Finalist / Top 20 placement in 5 additional national-level hackathons.

• Earned the AWS Solutions Architecture and Data Analytics certificates from Deloitte, and certified as a Google Gemini Certified Educator by Google.